

MOHAMMAD NOUR AL AWAD

Researcher & Machine Learning Engineer | PhD Student

Saint Petersburg, Russia | mohammad1999alawad@gmail.com

[LinkedIn](#) | [Telegram](#) | [Google Scholar](#) | [Scopus](#)

PROFESSIONAL SUMMARY

PhD researcher and machine learning engineer focused on adaptive Human-AI collaboration, coding agents, developer personalization, and efficient LLM systems. Lead applied research from problem formulation and architecture through large-scale experimentation, deployment, evaluation, and publication. Hands-on experience includes repository-level agents, context and memory, model routing, RAG, multimodal/web agents, backend systems, vLLM serving, Docker/Linux infrastructure, and MLOps.

CORE EXPERTISE

Research & Agentic Systems: Human-AI Collaboration; Agentic AI Systems; Coding Agents; LLM Systems; Repository-Level Agents; LLM-Assisted Software Engineering; Adaptive Context & Memory; Developer Personalization; Agent Evaluation & Benchmarking; Model Routing & Orchestration.

LLM & Agent Infrastructure: Multi-Model LLM Serving; vLLM; LiteLLM; RAG Systems; Tool-Using Agents; Sandboxed Agent Execution; Long-Context LLM Systems; LLM Evaluation Pipelines.

Data & Experimentation: Repository Mining; Developer Behavior Analytics; Benchmark Design; Large-Scale Data Collection; Experiment Design; Ablation Studies; Reproducible ML Experiments.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer / Researcher | ITMO University

Saint Petersburg, Russia | Mar 2024 – Present

- Lead the ITMO research and engineering work described below across Human-AI collaboration, AI developer tools, personalization, agentic systems, and LLM infrastructure, connecting research direction, architecture, orchestration, experimentation, implementation, deployment, evaluation, and publications.
- Led developer-behavior modeling and adaptive LLM code-assistance research covering intent inference, invocation timing, output filtering, and context-aware assistance; this work produced two IEEE/ACM ASE Workshops 2025 papers.
- Led the design and evaluation of repository-grounded coding-agent pipelines spanning exploration, file-level issue localization, repair, and validation. In Loc2Repair, predicted localization improved pooled SWE-bench Verified resolved rate from 44.7% to 49.1%, while gold localization reached 52.4%.
- Led research on cost-effective repository exploration across 999 benchmark tasks; lower-cost explorers retained about 78-94% of reference Hit@3 and 73-92% of F1 while reducing agent time by 41-88% and token use by 84-95%.
- Led and executed the transformation of Roo-Code into ITMO Code, a full-stack AI development platform for education with editor-native assistance, role-based workflows, Git/GitHub automation, sandboxed execution, agentic code review, backend APIs, workers, analytics, and automated pipelines.
- Led the design and development of Coding Agent Personalizer, a VS Code extension and service that learns from repository activity to propose reviewable coding-agent rules and reusable skills, with developer profiling and related-context workflows.

- Led deployment and operation of multi-model LLM serving infrastructure with vLLM and LiteLLM on multi-GPU NVIDIA RTX 6000 Ada servers, supporting coding, chat, and multimodal models while tuning context length, parallelism, reliability, and serving behavior.
- Led development of multimodal and web-agent systems for conversational data analysis and browser automation, combining sandboxed execution, tool use, multi-step reasoning, visualization, and multimodal interaction.
- Led repository-scale developer-behavior data collection and analysis covering 1.47M+ events, 521K+ commits, 123K+ pull requests, 220K+ PR review comments, 228K+ PR reviews, and 40K+ developers.

Software Engineer (C++/Python) | ViShare Technology

China | Jun 2023 - Apr 2024

- Developed and optimized algorithms for wireless video and HEVC products using C/C++ and Python, integrating team solutions into production-oriented streaming systems.
- Built profiling tools that improved analysis speed by 10x, automated testing workflows, and ported high-level synthesis components using Vitis HLS.

Machine Learning Researcher | Higher Institute for Applied Sciences and Technology (HIAST)

Syria | Jan 2022 - Sep 2022

- Led applied computer-vision R&D for edge deployment, including an automatic license-plate recognition system; improved model accuracy from 68% to 79%.
- Deployed detector applications on NPU-powered edge devices with IP-camera support and developed a Flutter application for live monitoring and archive browsing.

SELECTED RESEARCH & PUBLICATIONS

Cost-Effective Repository Exploration for Agentic Issue Localization — AgenticDev 2026, co-located with the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026), 2026. | [Paper](#)

Loc2Repair: A Framework for Evaluating the Impact of File-Level Issue Localization in Repo-Level LLM Repair — 2nd Generative Code Intelligence Workshop (GeCoIn 2026), co-located with IJCAI-ECAI 2026; CEUR Workshop Proceedings Vol. 4238, 2026. | [Paper](#)

A Multimodal Conversational Agent for Tabular Data Analysis — IEEE International Conference on Data Mining Workshops (ICDMW), BigIS Workshop, 2025. | [Paper](#)

Optimizing LLM Code Suggestions: Feedback-Driven Timing with Lightweight State Bounds — IEEE/ACM Automated Software Engineering Workshops (ASEW), Intelligent Software Engineering (ISE) Workshop, 2025. | [Paper](#)

Pre-Filtering Code Suggestions using Developer Behavioral Telemetry to Optimize LLM-Assisted Programming — IEEE/ACM Automated Software Engineering Workshops (ASEW), ASYDE Workshop, 2025. | [Paper](#)

SELECTED RESEARCH & ENGINEERING PROJECTS

Coding Agent Personalizer — VS Code extension and backend that turns developer/repository activity into reviewable agent rules, reusable skills, profiles, and related-context suggestions. | [Project](#)

ITMO Code — Led and executed the transformation of Roo-Code into a full-stack AI education platform with editor-native assistance, role-based orchestration, Git/GitHub automation, sandboxed execution, agentic review, backend services, analytics, and automated workflows. | [Project](#)

IssueLoc-Bench — Benchmark infrastructure for agentic file-level issue localization from real GitHub issues and fix snapshots, measuring quality together with steps, time, and token efficiency. | [Project](#)

Multi-Model LLM Serving Infrastructure — Led the design, deployment, and operation of a vLLM/LiteLLM serving stack for local coding, chat, and multimodal models on multi-GPU infrastructure, including long-context serving, tensor parallelism, routing, reliability, and operational debugging.

EDUCATION

PhD, Artificial Intelligence / Human-Centered AI | ITMO University

Saint Petersburg, Russia | 2025 - 2028 (Expected)

- Dissertation direction: Adaptive Human-AI Collaboration via Agentic Systems, with research on coding agents, context and memory, adaptive interaction, model orchestration, and agent evaluation.

Master's Degree, Big Data and Machine Learning | ITMO University

Saint Petersburg, Russia | 2023 – 2025

- Graduated with honors (Red Diploma), GPA 5/5, ranked #1.

Bachelor's Degree, Software Engineering | Higher Institute for Applied Sciences and Technology (HIAST)

Syria, Damascus | 2017 – 2022

- Valedictorian graduate award; excellence rate 86.17%.

High School Diploma, Distinguished Individuals Program | National Center for Distinguished (NCD)

Syria, Homs | 2017

- Completed Syria's selective national program for distinguished students, emphasizing mathematics, science, problem solving, independent learning, and research-oriented education.

ACADEMIC SERVICE

- Supervise 8 master's students at ITMO on coding agents, context and memory, model routing, RAG, and agent evaluation.
- Reviewer, EMNLP 2025 Industry Track.
- Presented a research poster on developer-personalized coding agents at Huawei Lake Baikal Summit 2026.

Languages: Arabic — Native; English — C2; Russian — B2.